

---

# Aim Short to Reach Far: Your Frozen World Model Can Plan Better Than You Think

---

Xvyuan Liu<sup>1</sup>, Jianjie Fang<sup>1</sup>, Chen Gao<sup>1,\*</sup>, Yong Li<sup>1,\*</sup>

<sup>1</sup>Tsinghua University

\*Corresponding authors: [chgao96@gmail.com](mailto:chgao96@gmail.com), [liyong07@tsinghua.edu.cn](mailto:liyong07@tsinghua.edu.cn)

 [Project Page](#)  [Code](#)

## Abstract

Planners built on visual world models commonly score each predicted outcome by its distance to the encoded goal image. We show that this target can limit control even with exact dynamics and globally optimal short-horizon search: reaching a goal may require actions that initially move away from it. With frozen LeWM models, intermediate targets substantially improve action synthesis and recorded-action ranking on Cube, PushT, Reacher, and TwoRoom. Learned targets and targets drawn from observed experience both produce these gains. We introduce *Anchored Planning*, which retrieves a recorded segment whose start and end resemble the current and goal observations, then aims at an observation shortly after its start. The frozen model scores actions toward this target from the current state. Without additional training, planning toward observed targets outperforms the released LeWM planner on every task in our long-range evaluation. Additional final-goal search falls short of the same gains. Lower successor-prediction error need not translate into better control. Success also depends on how far ahead the target is placed and on shrinking the retrieval span as execution advances. Changing only the target lets the same frozen model and planner reach goals that final-goal scoring misses.

## 1 Introduction

Visual world models predict how observations change under candidate actions [Zhou et al., 2025, Assran et al., 2025, Maes et al., 2026]. A common planner selects the predicted endpoint closest to the encoded goal image, executes a short block, and replans [Ebert et al., 2018, Hafner et al., 2019]. This couples two choices that objective-driven architectures assign to separate modules [LeCun, 2022]: how to predict an action’s effect and which outcome to pursue. We show that the second choice alone can decide whether the same frozen predictor and planner reach the goal.

Reaching a goal can require first moving farther from it. Classical potential-field methods encounter this difficulty when local descent leads into a nonglobal minimum [Khatib, 1986, Rimon and Koditschek, 1992]. Short-horizon distance scoring inherits the same *objective mismatch*: it can reject the first steps of a useful route before seeing where they lead. Proposition 1 makes this precise. In a smooth control problem, exact predictions and globally optimal short-horizon search still select inaction forever, whereas successive observed targets on the same feasible route lead to the goal. Prediction and search can therefore be correct while the objective favors the wrong action.

With frozen LeWM predictors, learned and observed intermediate targets both substantially improve action synthesis and recorded-action selection. Observed targets have a larger mean advantage over final-goal scoring at longer goal offsets. At the 100–150-action offsets, short observed-target searches also exceed the released LeWM on every task. Running more final-goal CEM iterations does not close this gap. Figure 1 illustrates how changing the destination used to score actions improves control with the same frozen dynamics.

Preprint.



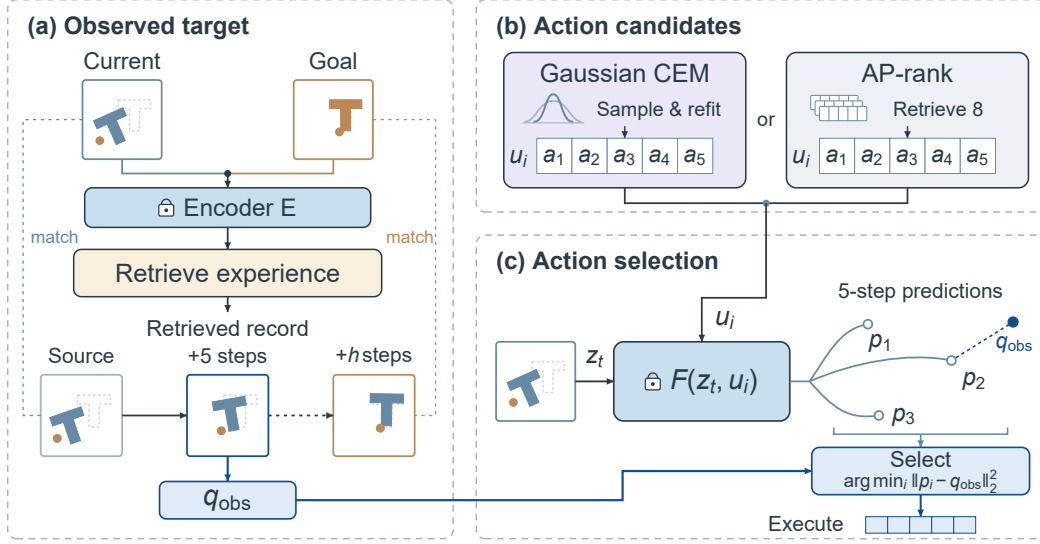


Figure 2: **Anchored Planning: observed targets, live predictions.** (a) We match the current and goal images to the source and distant images of each record. The closest record’s five-step successor supplies  $q_{\text{obs}}$ . The distant offset is  $h = \max(5, H - t)$ , for goal offset  $H$  and  $t$  executed actions. (b) Each strip represents a candidate block  $u_i$ . Gaussian CEM synthesizes these blocks, whereas AP-rank retrieves eight recorded blocks. (c) The frozen model predicts endpoints  $p_i = F(z_t, u_i)$ . The dashed segment shows distance to  $q_{\text{obs}}$ . We select the block whose endpoint is closest to the target, execute up to five actions, and observe again before replanning. All glyphs and positions are schematic.

For two fixed predictions  $p_u = F(z_t, u)$  and  $p_v = F(z_t, v)$ , let  $\Delta_q(u, v) = J_q(u) - J_q(v)$ . A negative value favors  $u$  over  $v$ . Changing the target from  $q$  to  $q'$  gives

$$\Delta_{q'}(u, v) - \Delta_q(u, v) = 2(p_u - p_v)^\top (q - q'). \quad (2)$$

Thus the same predicted endpoints can favor different actions when the target changes. The following construction shows how this changes control even when the short-horizon search finds the global minimum.

**Proposition 1** (Goal-distance control can stall). *Even with smooth dynamics, bounded continuous actions, and exact predictions, globally optimal five-action final-goal scoring can select the zero block forever from a continuum of goal-reachable states. Under identical dynamics, actions, and execution, six successive endpoints of one feasible recorded trajectory instead lead every such start to the goal.*

The construction places states on the curve  $c(s) = (s, s^2)$ , with goal  $c(2)$ . Each primitive action increases  $s$  by an amount in  $[0, 1/8]$ , so exact predictions follow the same continuous route toward the goal. Yet squared final-goal distance along that route is

$$D(s) = \|c(s) - c(2)\|^2 = (s - 2)^2 + (s^2 - 4)^2. \quad (3)$$

For starts  $s \in [-3/2, -1]$ , every nonzero admissible block ends farther from the goal under  $D$ , whereas the zero block leaves the distance unchanged. The global optimum of the local objective is therefore to stay still. The predictor correctly places every nonzero block farther along the route, but the distance score favors staying still.

Replacing the final goal with successive targets on the recorded curve changes this choice. Each target can be attained within the local action horizon in this construction, and selecting it advances along the same dynamics until the goal is reached (see Appendix B for the complete construction and proof). Anchored Planning applies this idea by retrieving a task-relevant trajectory and predicting actions toward an early observed successor. Section 4 measures the resulting target effect with frozen visual predictors.

### 3 Anchored Planning

Figure 2 shows how Anchored Planning separates choosing a destination from choosing an action. We take an observed target from recorded experience and use the frozen model to predict how candidate actions approach it from the current state. We consider two memory settings. *Observation-only memory* contains ordered images with episode and time indices but no actions, so the planner must synthesize them. *Action-labeled memory* also contains the recorded actions, which the planner can rank. Both use the same frozen predictor, target rule, and fixed memory.

#### 3.1 Use distant experience to choose an observed target

A short search needs a nearby destination that belongs to a route toward the final goal. We therefore match a record through its source and distant observations, then use its early successor to score actions. Each record contains a source, an  $L$ -step successor, and an  $h$ -step endpoint. We use the distant endpoint to match the longer route to the goal, while predicting only the next action block.

Each query provides the recorded action offset  $H$  between its starting image and goal image. After  $t$  executed primitive actions, we retrieve records spanning

$$h(t) = \max\{L, H - t\}. \quad (4)$$

As execution advances, retrieval matches progressively shorter recorded spans to the final goal, down to  $L$ . The observed target and prediction horizon remain  $L$  steps ahead of their respective source states.

Let  $s$  index a recorded start. For retrieval horizon  $h$ , memory  $\mathcal{B}_h$  contains starts for which both  $s + L$  and  $s + h$  remain within the same episode. Each record pairs the retrieval key

$$b_s^{(h)} = (z_s, z_{s+h}, z_{s+h} - z_s)$$

with its successor encoding  $z_{s+L}$ . Retrieval uses ordered observations and episode indices. For AP-rank, a record additionally provides its next-five-action block  $u_s = (a_s, \dots, a_{s+L-1})$ . The current query is  $b_t = (z_t, z_g, z_g - z_t)$ . Retrieval ranks records by standardized squared distance,

$$d_h(t, s) = \left\| \frac{b_t - b_s^{(h)}}{\sigma_h} \right\|_2^2. \quad (5)$$

Here  $\sigma_h$  is the coordinate-wise population standard deviation over eligible records. The closest start  $s_1$  supplies the target

$$q_{\text{obs}} = z_{s_1+L} = E(o_{s_1+L}). \quad (6)$$

#### 3.2 Synthesize actions from observation-only memory

AP-CEM maintains a Gaussian distribution over five-action blocks. From a fixed prior, it clips proposals to the action bounds and scores their predicted endpoints against  $q_{\text{obs}}$ . The best-scoring proposals update the distribution. We compare the clipped means before and after optimization and execute the lower-cost block. Appendix A gives the complete update and action mapping.

#### 3.3 Rank retrieved actions from the current observation

AP-rank takes one action block from each of the eight retrieved records and predicts its endpoint from the current observation. We score all eight endpoints against the same observed target, taken from the closest record. Let  $P$  clip each action coordinate to its task-specific bounds. The clipped candidate set is  $\mathcal{C} = \{P(u_{s_1}), \dots, P(u_{s_8})\}$ , and AP-rank selects

$$\hat{u} = \arg \min_{u \in \mathcal{C}} \|F(z_t, u) - q_{\text{obs}}\|_2^2, \quad (7)$$

with ties resolved by retrieval order. After each primitive action, we check for success, termination, and the end of the action allowance. We observe and replan after at most five actions (Algorithm 1 in Appendix A.6).

#### 3.4 Compared targets and baselines

We compare final-goal, learned, and observed targets under each action rule. Final-goal scoring sets  $q = z_g$ . A learned-target MLP predicts the observed successor from the current and goal encodings, their difference, and the retrieval span. It is trained on the same memory observations as AP, and its configuration is selected on held-out memory episodes (Appendix C.1).

Table 1: **Intermediate targets improve control with the same frozen model.** We report success (%) on 128 paired queries per task. Column groups indicate what experience each controller can use. Learned targets are trained on memory observations. For perturbed starts, we average within each query, and task means weight all tasks equally. Bold marks the highest success within each group of controllers. AP-CEM and AP-rank use observed targets.

| Task    | Start     | No memory | Observation-only |             |             | Recorded actions |            |              |              |
|---------|-----------|-----------|------------------|-------------|-------------|------------------|------------|--------------|--------------|
|         |           | LeWM      | CEM final        | CEM learned | AP-CEM      | Direct           | Rank final | Rank learned | AP-rank      |
| Cube    | Standard  | 15.6      | 7.0              | <b>62.5</b> | 46.9        | 82.0             | 62.5       | <b>91.4</b>  | 73.4         |
|         | Perturbed | 12.5      | 2.7              | <b>38.7</b> | 29.7        | 47.7             | 41.8       | <b>50.0</b>  | 45.7         |
| PushT   | Standard  | 7.0       | 2.3              | 52.3        | <b>69.5</b> | 51.6             | 25.8       | 54.7         | <b>66.4</b>  |
|         | Perturbed | 9.4       | 2.7              | 48.8        | <b>64.8</b> | 18.0             | 12.9       | 35.5         | <b>47.7</b>  |
| Reacher | Standard  | 66.4      | 25.8             | 54.7        | <b>75.8</b> | <b>99.2</b>      | 90.6       | 95.3         | 96.9         |
|         | Perturbed | 68.8      | 22.7             | 59.0        | <b>78.9</b> | <b>96.9</b>      | 87.5       | <b>96.9</b>  | 95.7         |
| TwoRoom | Standard  | 17.2      | 1.6              | <b>53.1</b> | 46.9        | <b>99.2</b>      | 89.1       | 96.9         | 98.4         |
|         | Perturbed | 16.8      | 1.6              | <b>55.1</b> | 46.9        | <b>100.0</b>     | 88.7       | 98.4         | <b>100.0</b> |
| Mean    | Standard  | 26.6      | 9.2              | 55.7        | <b>59.8</b> | 83.0             | 67.0       | <b>84.6</b>  | 83.8         |
|         | Perturbed | 26.9      | 7.4              | 50.4        | <b>55.1</b> | 65.6             | 57.7       | 70.2         | <b>72.3</b>  |

For CEM, only the target changes: it determines which predicted outcomes refine the Gaussian, while initialization, seeds, search settings, and action mapping stay fixed. Ranking scores the same eight retrieved blocks from a shared live state. Each controller then repeats its target rule and action selection along the resulting rollout.

*Direct* executes the closest record’s clipped action block, then observes and retrieves again. Comparing Direct with predictive ranking measures the benefit of evaluating recorded actions from the current state.

## 4 Experiments

We organize the experiments around three questions about the effect of target choice, the role of search, and the design of useful targets:

**RQ1:** How much control does changing the target reveal?

**RQ2:** Can more search or longer lookahead recover the same gains?

**RQ3:** What makes a target useful, and how does prediction help reach it?

### 4.1 Evaluation setup

**Models, memory, and paired queries.** We evaluate Cube, PushT, Reacher, and TwoRoom with task-specific pretrained LeWM models [Maes et al., 2026]. Encoders, predictors, and the original action normalizers remain frozen. Memory contains the supplied training trajectories. Each task has 128 query episodes. We exclude them from memory and never draw perturbation prefixes from them. Each query pairs a start image with the image  $H$  actions later in the same episode. We call these the original offsets, which span 100–150 actions (Table 5). All experiments below use these paired queries. Appendix F repeats the target comparison on independent queries.

**Starts and success.** Recorded actions can have different effects from a displaced state. We therefore evaluate each recorded (standard) start and two perturbed starts reached by executing preassigned five-action prefixes. The goal, memory, and subsequent allowance stay fixed. Controllers receive images,  $H$ , and the allowance. We use physical states to restore paired starts and measure success after every primitive action. For perturbed starts, we average the two starts of each query, and task means weight all tasks equally. Appendix A gives the evaluation details.

**Action rules and targets.** Single-block CEM runs 30 iterations with 300 candidates and 30 elites, and ranking chooses among eight retrieved blocks. Both use the final, learned, or observed targets defined in Section 3.4. They replan after at most five actions and clip proposals to the action bounds before prediction and execution. LeWM keeps its released action mapping and plans and executes 25 actions at a time. On shared queries, our implementation and the released evaluator agree on nearly all episode outcomes (Appendix D.2).

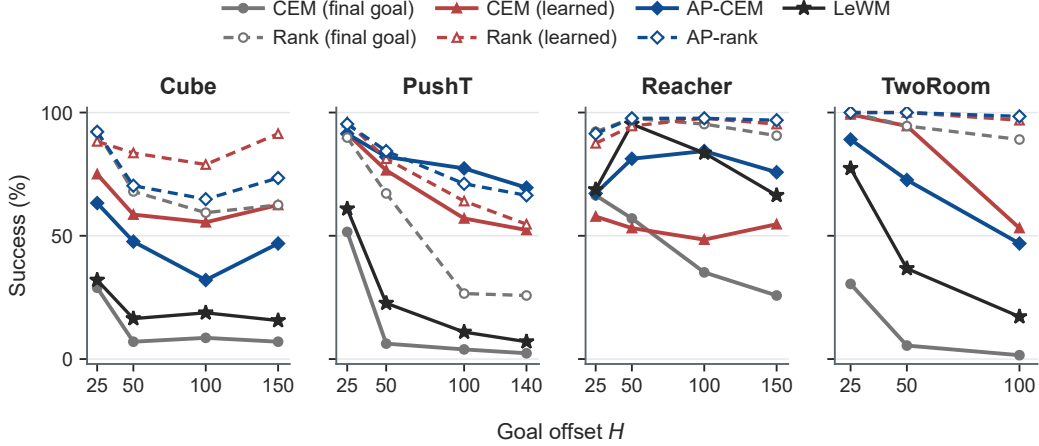


Figure 3: **Observed targets give larger mean gains beyond the shortest offset.** We keep the standard starts fixed as  $H$  changes and adjust the action allowance to preserve each task’s  $B/H$ . Planner lookahead stays fixed. Results at the original offsets are those in Table 1. Table 12 gives perturbed results.

#### 4.2 RQ1: How much control does changing the target reveal?

**Both intermediate targets improve control.** Table 1 shows that changing the target while retaining the frozen predictor and action rule improves success on every task at standard and perturbed starts. At standard starts, CEM’s mean rises from 9.2% with the final goal to 55.7% with a learned target and 59.8% with an observed target. Ranking rises from 67.0% to 84.6% and 83.8%, respectively. Thus both learned and observed targets let the same frozen model and action rule succeed where final-goal scoring fails. AP achieves these gains with targets drawn directly from recorded experience.

**Target choice matters more at longer offsets.** Figure 3 compares goal offsets while keeping starts and lookahead fixed. Final-goal CEM declines from the shortest to the original offset on all tasks. Caselli et al. [2026] observe the same decline for the non-hierarchical LeWM planner on PushT, with success rates of 94.0%, 52.7%, and 18.0% at offsets 25, 50, and 75 under their protocol. At standard starts, as the goal offset grows from  $H = 25$  to  $H = 100$ , the mean gain from observed targets rises from 33.4 to 47.9 percentage points for CEM and from 1.2 to 15.4 points for ranking. Perturbed starts show the same trend (Table 12). When the goal is nearby, ranking with the final goal already works well, and intermediate targets matter mainly once the goal is farther away.

**The recovered control includes physical detours.** We normalize physical error by the task’s success thresholds. A success counts as a detour when this error rises above a previous minimum by a fixed margin, and a failure counts as stalling when there is little task-relevant motion near the end of the rollout. Successful actions can initially increase physical error, as the construction in Section 2 illustrates. Figure 4 shows this on PushT: at standard starts, all 89 AP-CEM successes take a detour, whereas 100 of 125 final-goal CEM failures stall.

#### 4.3 RQ2: Can more search or longer lookahead recover the same gains?

**Target changes outperform more final-goal search.** RQ1 kept the search budget fixed. Here we ask whether more search or a longer lookahead can close the same gap. Figure 5 shows that two CEM iterations with either intermediate target already outperform 30 iterations with the final goal, on every task and at both kinds of start.

**Short observed-target search exceeds LeWM at the original offsets.** AP-CEM outperforms LeWM on every task and at both kinds of start. It uses 63.4–85.5% as many predicted action blocks per episode as LeWM on Cube, PushT, and TwoRoom, and 126.6–128.1% as many on Reacher. As Figure 3 shows, LeWM leads at some shorter Reacher offsets. When LeWM replans every five actions, it still falls below AP-CEM on every task and at both kinds of start (Table 11).

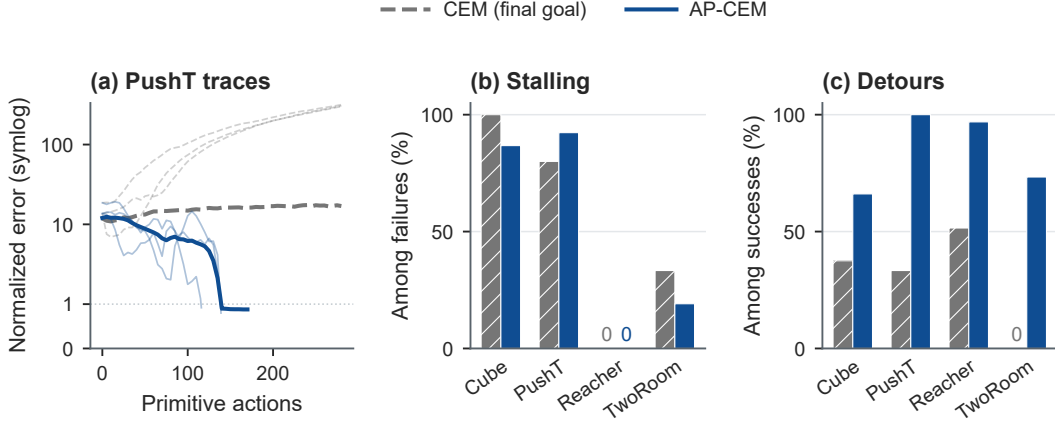


Figure 4: **Successful PushT control takes detours while final-goal failures often stall.** (a) We compare final-goal CEM failures with AP-CEM successes at standard starts. Thin curves show the first three eligible queries, and thick curves show group medians with terminal error carried forward. The axis is linear below one and logarithmic above. (b) Bars show the fraction of failures that stall. (c) Bars show the fraction of successes that take a detour. Both rates use rollouts that enter control. Appendix C.2 gives counts and threshold sensitivity.

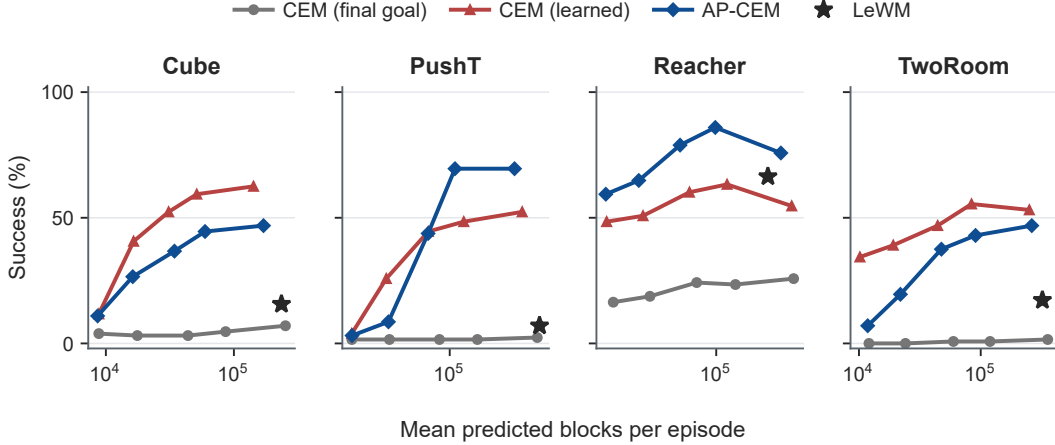


Figure 5: **Two CEM iterations with an intermediate target outperform 30 with the final goal on every task.** Curves show standard-start success at iteration counts 1, 2, 5, 10, 30, and stars mark LeWM. We measure work by the mean number of five-action blocks predicted per episode, counted until the episode ends. The same comparison holds at perturbed starts (Table 9).

#### 4.4 RQ3: What makes a target useful, and how does prediction help reach it?

**Successor accuracy and control favor different target choices.** The learned target is trained to match a recorded successor, whereas control is evaluated by reaching the final goal. We first compare target accuracy with control success. Table 2 shows that the learned target predicts held-out successors with lower mean error in every task, while observed targets achieve higher mean CEM success. The control advantage varies across tasks: learned targets do better on Cube and TwoRoom, and observed targets do better on PushT and Reacher. Lower successor error therefore does not consistently predict higher control success.

**Nearby targets and retrieval timing shape success.** Table 3 compares target offsets while keeping predictions at five actions. A three-step target raises AP-CEM’s mean success to 69.9%, while AP-rank performs best at the default five-step offset among those tested. One- and ten-step targets give lower means than the default for both rules. Ranking depends on shrinking the retrieval span as execution advances. With the span fixed at  $H$ , standard-start success falls from 83.8% to 9.6%. With the adaptive span, using ten percent of memory preserves most of the success rate. Dropping the distant endpoint from the retrieval key slightly raises the ranking mean (Table 3b).

Table 2: **Lower successor error does not always mean higher control success.** We measure the mean squared distance to the recorded successor on held-out memory transitions and CEM success on standard-start evaluation queries. Bold marks higher success. Appendix C.1 explains how target accuracy is measured.

| Task    | Successor error |          | CEM success (%) |             |
|---------|-----------------|----------|-----------------|-------------|
|         | Learned         | Observed | Learned         | Observed    |
| Cube    | 19.708          | 63.056   | <b>62.5</b>     | 46.9        |
| PushT   | 3.774           | 10.668   | 52.3            | <b>69.5</b> |
| Reacher | 74.724          | 154.351  | 54.7            | <b>75.8</b> |
| TwoRoom | 133.857         | 266.676  | <b>53.1</b>     | 46.9        |

Table 3: **Target placement and retrieval timing determine how memory guides control.** We average success (%) across the four tasks. Predictions span  $L = 5$  actions, and  $\ell$  sets the target’s offset in the recorded trajectory. When varying memory size, we keep the default target offset. In the fixed-span setting, we use  $h = H$  throughout execution. Bold marks the highest mean per column within each panel.

(a) Target offset and memory size

| Variant                     | AP-rank     |             | AP-CEM      |             |
|-----------------------------|-------------|-------------|-------------|-------------|
|                             | Standard    | Perturbed   | Standard    | Perturbed   |
| Target $\ell = 1$           | 69.5        | 61.3        | 43.0        | 37.9        |
| Target $\ell = 3$           | 78.5        | 68.5        | <b>69.9</b> | <b>62.6</b> |
| Target $\ell = 5$ (default) | <b>83.8</b> | <b>72.3</b> | 59.8        | 55.1        |
| Target $\ell = 10$          | 81.6        | 68.4        | 54.1        | 52.7        |
| Memory 10%                  | 77.5        | 68.9        | 56.6        | 52.1        |
| Memory 25%                  | 80.1        | 70.4        | 58.4        | 53.5        |
| Memory 50%                  | 82.0        | 71.0        | 58.6        | 53.8        |

(b) Retrieval design for AP-rank

| Variant                      | Standard    | Perturbed   |
|------------------------------|-------------|-------------|
| Default                      | 83.8        | 72.3        |
| Key without distant endpoint | <b>85.7</b> | <b>72.9</b> |
| Key without displacement     | 83.0        | 72.1        |
| Fixed retrieval span         | 9.6         | 11.3        |

**An observed destination adjusts the requested motion.** For recorded displacement  $d = z_{s+L} - z_s$  and live offset  $e = z_t - z_s$ , the transported target  $z_t + d$  requests the full recorded motion, whereas the observed target requests  $d - e$ . Anchoring the target to the observed endpoint raises mean CEM success from 53.7% to 59.8% at standard starts and from 46.0% to 55.1% after perturbation. Table 16 breaks this down by task: anchoring gives higher success on Cube, PushT, and TwoRoom, while transport does better on Reacher.

**Live prediction improves the choice of action.** To assess how actions approach the target, we execute all eight candidates in the simulator from the same start and compare the resulting endpoints with estimates from displacement transfer and the frozen predictor, respectively:

$$\tilde{p}_j = z_t + (z_{s_j+L} - z_{s_j}), \quad p_j = F(z_t, u_j). \quad (8)$$

Table 4 shows that the predictor reduces both endpoint error and selection regret relative to displacement in every task and start setting. It also reduces regret relative to Direct. We measure error as squared distance to the simulated endpoint and regret as the chosen action’s exact observed-target cost above the best candidate’s cost. Appendix E.3 gives first-block interventions followed by Direct in every rollout.

In closed-loop control, AP-rank gains most over Direct on PushT, raising success from 51.6% to 66.4% at standard starts and from 18.0% to 47.7% after perturbation. Experience specifies where to aim, and live prediction determines which recorded action approaches it from the current state.

## 5 Related Work

**Planning objectives.** Artificial potential fields turn goals and obstacles into control objectives [Khatib, 1986]. With known free-space geometry, navigation functions can remove nonglobal minima [Rimon and Koditschek, 1992]. Latent planners score predicted features [Sobal et al., 2025],

Table 4: **Live-state prediction improves endpoint accuracy and action selection.** We report mean squared latent distances in task-specific units, with lower values indicating better performance. Bold marks each metric’s minimum. Displacement transfers recorded motion, whereas Direct executes the closest record’s action. We exclude starts if any candidate ends before five actions (at most 3 of 128 standard and 6 of 256 perturbed starts per task).

| Task    | Start     | Endpoint error |         | Selection regret |        |        |
|---------|-----------|----------------|---------|------------------|--------|--------|
|         |           | Predictor      | Displ.  | Predictor        | Displ. | Direct |
| Cube    | Standard  | <b>7.538</b>   | 35.344  | <b>5.332</b>     | 7.535  | 11.375 |
|         | Perturbed | <b>37.726</b>  | 66.860  | <b>8.472</b>     | 11.731 | 14.903 |
| PushT   | Standard  | <b>1.532</b>   | 3.078   | <b>0.200</b>     | 0.580  | 0.849  |
|         | Perturbed | <b>2.980</b>   | 9.562   | <b>0.753</b>     | 2.418  | 4.449  |
| Reacher | Standard  | <b>25.243</b>  | 51.917  | <b>8.644</b>     | 15.058 | 18.640 |
|         | Perturbed | <b>25.587</b>  | 50.862  | <b>7.401</b>     | 12.558 | 16.296 |
| TwoRoom | Standard  | <b>13.560</b>  | 162.035 | <b>1.433</b>     | 29.838 | 41.749 |
|         | Perturbed | <b>12.532</b>  | 160.497 | <b>1.283</b>     | 20.154 | 37.307 |

while Universal Planning Networks and Monotone Cost Ranking learn representations or costs for control [Srinivas et al., 2018, Chahe et al., 2026]. Terver et al. [2026] study optimizers and latent costs, showing that accurate prediction need not yield successful planning. Concurrent studies test decoded-position and learned temporal costs on TwoRoom [Singh, 2026] or learn costs over latent rollouts [Huang et al., 2026]. We isolate target-induced stalling with exact dynamics and optimal short-horizon search, then compare targets with a fixed learned predictor, metric, and search.

**Recorded and generated intermediate targets.** SPTM and SoRB search graphs of observations using learned connectivity or distances [Savinov et al., 2018, Eysenbach et al., 2019]. PALMER likewise uses learned reachability to stitch recorded segments [Beker et al., 2022]. HWM predicts subgoals with a high-level world model [Zhang et al., 2026], and Hi-LeWM improves subgoal quality by constraining high-level search near encoded training sequences [Caselli et al., 2026]. SAGE learns target and action generators with frozen LeWM, using expert trajectories and low-dimensional state [Cheng et al., 2026]. Its generator-only variant is closest to our observation-only CEM because it scores CEM candidates against a learned target. AP instead takes an observed successor from a retrieved segment as its target. Our learned-target baseline compares successor prediction with retrieval using the same visual inputs and memory.

**Experience for action and target choice.** VINN predicts actions by local regression on visual neighbors [Pari et al., 2022]. Our Direct baseline tests immediate action reuse under our current-goal retrieval rule. Recorded experience has also been used to estimate transitions [Malato and Hautamäki, 2025], to aid hybrid LeWM planning [Gao et al., 2026], to learn policies and trajectory models [Florence et al., 2022, Chi et al., 2023, Janner et al., 2022], to relabel goals in hindsight [Andrychowicz et al., 2017], and to learn rewards from video [Chen et al., 2021]. OGBench provides offline goal-conditioned benchmarks built from such data [Park et al., 2025]. In AP, we use experience to choose the destination and live-state prediction to decide which synthesized or retrieved action approaches it.

## 6 Conclusion

Changing the target can let the same frozen predictor and planner reach goals that final-goal scoring misses. Our construction shows that final-goal scoring can fail even with exact predictions and globally optimal short-horizon search. On the LeWM tasks, learned and observed intermediate targets achieve higher success than additional final-goal search. Goal-distance planning success, a common world-model benchmark, thus reflects the objective as well as the predictor. A nearby target, a retrieval span that shrinks during execution, and prediction from the live state all contribute to control. Testing the same target choices with other world-model families could show how the gains depend on the predictor.

## References

- Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL <https://papers.nips.cc/paper/2017/hash/453fadbd8a1a3af50a9df4df899537b5-Abstract.html>. 9
- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023. doi: 10.1109/cvpr52729.2023.01499. URL [https://openaccess.thecvf.com/content/CVPR2023/html/Assran\\_Self-Supervised\\_Learning\\_From\\_Images\\_With\\_a\\_Joint-Embedding\\_Predictive\\_Architecture\\_CVPR\\_2023\\_paper.html](https://openaccess.thecvf.com/content/CVPR2023/html/Assran_Self-Supervised_Learning_From_Images_With_a_Joint-Embedding_Predictive_Architecture_CVPR_2023_paper.html). 14
- Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhohus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khilodov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025. URL <https://arxiv.org/abs/2506.09985>. 1
- Randall Balestriero and Yann LeCun. LeJEPa: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025. URL <https://arxiv.org/abs/2511.08544>. 14
- Onur Beker, Mohammad Mohammadi, and Amir Zamir. PALMER: Perception–action loop with memory for long-horizon planning. In *Advances in Neural Information Processing Systems*, volume 35, 2022. doi: 10.52202/068431-2483. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/hash/dd7a48c862b800f0537fe1d506e641b5-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2022/hash/dd7a48c862b800f0537fe1d506e641b5-Abstract-Conference.html). 9
- Niccolò Caselli, Francesco Massafra, Samuele Punzo, Salvatore Lo Sardo, Ippokratis Pantelidis, and Sathya Kamesh Bhethanabhotla. Mind the gap: Promises and pitfalls of hierarchical planning in LeWorldModel. *arXiv preprint arXiv:2607.12547*, 2026. URL <https://arxiv.org/abs/2607.12547>. 6, 9
- Amirhosein Chahe, Siwei Cai, and Lifeng Zhou. Latent world models with monotone planning costs for image-goal navigation. *arXiv preprint arXiv:2608.09073*, 2026. doi: 10.48550/arXiv.2608.09073. URL <https://arxiv.org/abs/2608.09073>. 9
- Annie S. Chen, Suraj Nair, and Chelsea Finn. Learning generalizable robotic reward functions from “in-the-wild” human videos. In *Proceedings of Robotics: Science and Systems*, 2021. doi: 10.15607/RSS.2021.XVII.012. URL <https://www.roboticsproceedings.org/rss17/p012.html>. 9
- Letian Cheng, Qi Zhang, and Yisen Wang. SAGE: Subgoal-conditioned action generation for latent world model planning. *arXiv preprint arXiv:2607.17973*, 2026. URL <https://arxiv.org/abs/2607.17973>. 2, 9
- Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.026. URL <https://www.roboticsproceedings.org/rss19/p026.html>. 9
- Pieter-Tjerk de Boer, Dirk P. Kroese, Shie Mannor, and Reuven Y. Rubinstein. A tutorial on the cross-entropy method. *Annals of Operations Research*, 134(1):19–67, 2005. doi: 10.1007/s10479-005-5724-z. URL <https://doi.org/10.1007/s10479-005-5724-z>. 2
- Frederik Ebert, Chelsea Finn, Sudeep Dasari, Annie Xie, Alex Lee, and Sergey Levine. Visual foresight: Model-based deep reinforcement learning for vision-based robotic control. *arXiv preprint arXiv:1812.00568*, 2018. URL <https://arxiv.org/abs/1812.00568>. 1

- Benjamin Eysenbach, Ruslan Salakhutdinov, and Sergey Levine. Search on the replay buffer: Bridging planning and reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/5c48ff18e0a47baaf81d8b8ea51eec92-Abstract.html>. 9
- Pete Florence, Corey Lynch, Andy Zeng, Oscar A. Ramirez, Ayzaan Wahid, Laura Downs, Adrian Wong, Johnny Lee, Igor Mordatch, and Jonathan Tompson. Implicit behavioral cloning. In *Proceedings of the 5th Conference on Robot Learning*, volume 164 of *Proceedings of Machine Learning Research*, pages 158–168. PMLR, 2022. URL <https://proceedings.mlr.press/v164/florence22a.html>. 9
- Baoqi Gao, Ruize Han, Miao Wang, and Song Wang. IMWM: Intuition models complement world models for latent planning. *arXiv preprint arXiv:2606.01626*, 2026. doi: 10.48550/arXiv.2606.01626. URL <https://arxiv.org/abs/2606.01626>. 9
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2555–2565. PMLR, 2019. URL <https://proceedings.mlr.press/v97/hafner19a.html>. 1
- Nicklas Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2310.16828>. 2
- Nicklas A. Hansen, Hao Su, and Xiaolong Wang. Temporal difference learning for model predictive control. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 8387–8406. PMLR, 2022. URL <https://proceedings.mlr.press/v162/hansen22a.html>. 2
- Xiaodi Huang, Ziyi Ding, Jingtian Wan, Yuchen Liu, Yuan Zhang, Xiao-Ping Zhang, Jiayu Chen, Zhang Zhang, and Tao Huang. Traj-LeWM: Path-aware world-model planning via latent trajectory cost. *arXiv preprint arXiv:2608.14125*, 2026. URL <https://arxiv.org/abs/2608.14125>. 9
- Michael Janner, Yilun Du, Joshua B. Tenenbaum, and Sergey Levine. Planning with diffusion for flexible behavior synthesis. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 9902–9915, 2022. URL <https://proceedings.mlr.press/v162/janner22a.html>. 9
- Oussama Khatib. Real-time obstacle avoidance for manipulators and mobile robots. *The International Journal of Robotics Research*, 5(1):90–98, 1986. doi: 10.1177/027836498600500106. 1, 8
- Yann LeCun. A path towards autonomous machine intelligence. OpenReview, 2022. URL <https://openreview.net/forum?id=BZ5a1r-kVsf>. 1
- Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorld-Model: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026. URL <https://arxiv.org/abs/2603.19312>. 1, 5, 13, 14, 19, 20
- Federico Malato and Ville Hautamäki. Zero-shot world models via search in memory. In *Advances in Neural Information Processing Systems*, 2025. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/file/da7d7341051b4e0444cc96c78f1e4df9-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2025/file/da7d7341051b4e0444cc96c78f1e4df9-Paper-Conference.pdf). 9
- Jyothish Pari, Nur Muhammad (Mahi) Shafiullah, Sridhar Pandian Arunachalam, and Lerrel Pinto. The surprising effectiveness of representation learning for visual imitation. In *Proceedings of Robotics: Science and Systems*, 2022. doi: 10.15607/RSS.2022.XVIII.010. URL <https://www.roboticsproceedings.org/rss18/p010.html>. 9

- Seohong Park, Kevin Frans, Benjamin Eysenbach, and Sergey Levine. OGBench: Benchmarking offline goal-conditioned RL. In *International Conference on Learning Representations*, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/ecd92623ac899357312aaa8915853699-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/ecd92623ac899357312aaa8915853699-Abstract-Conference.html). 9
- Elon Rimon and Daniel E. Koditschek. Exact robot navigation using artificial potential functions. *IEEE Transactions on Robotics and Automation*, 8(5):501–518, 1992. doi: 10.1109/70.163777. 1, 8
- Nikolay Savinov, Alexey Dosovitskiy, and Vladlen Koltun. Semi-parametric topological memory for navigation. In *International Conference on Learning Representations*, 2018. URL <https://openreview.net/forum?id=SygwwGbRW>. 9
- Joyjeet Singh. The objective is the bottleneck: Latent world models encode what their planners cannot use. *arXiv preprint arXiv:2608.12959*, 2026. URL <https://arxiv.org/abs/2608.12959>. 9
- Uladzislau Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestrieri, Tim G. J. Rudner, and Yann LeCun. Learning from reward-free offline data: A case for planning with latent dynamics models. In *Advances in Neural Information Processing Systems*, volume 38, pages 43905–43941, 2025. doi: 10.52202/085713-1465. URL [https://proceedings.neurips.cc/paper\\_files/paper/2025/hash/3e7cf447f21cd11c846463affefce665-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2025/hash/3e7cf447f21cd11c846463affefce665-Abstract-Conference.html). 8
- Aravind Srinivas, Allan Jabri, Pieter Abbeel, Sergey Levine, and Chelsea Finn. Universal planning networks: Learning generalizable representations for visuomotor control. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 4732–4741, 2018. URL <https://proceedings.mlr.press/v80/srinivas18b.html>. 9
- Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models? *Transactions on Machine Learning Research*, 2026. URL <https://openreview.net/forum?id=chZn5Gdh8e>. 9
- Wancong Zhang, Basile Terver, Artem Zhohus, Soham Chitnis, Harsh Sutaria, Mido Assran, Randall Balestrieri, Amir Bar, Adrien Bardes, Yann LeCun, and Nicolas Ballas. Hierarchical planning with latent world models. *arXiv preprint arXiv:2604.03208*, 2026. URL <https://arxiv.org/abs/2604.03208>. 2, 9
- Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 79115–79135, 2025. URL <https://proceedings.mlr.press/v267/zhou25t.html>. 1

The appendix follows the main questions: evaluation protocol and proofs first, then target comparisons and behavior (RQ1), search and goal distance (RQ2), and target construction and action selection (RQ3). The final section reports an independent-query replication. Code and per-episode results are available in our [repository](#).

## A Evaluation Protocol

Table 5: **Evaluation inputs.**  $H$  is the recorded goal offset,  $B$  the primitive-action allowance, and memory the number of eligible episodes.

| Task   | Cube  | PushT  | Reacher | TwoRoom |
|--------|-------|--------|---------|---------|
| $H$    | 150   | 140    | 150     | 100     |
| $B$    | 150   | 280    | 300     | 200     |
| Memory | 8,000 | 16,688 | 8,000   | 8,000   |

### A.1 Query construction and retrieval memory

Memory consists of the offline training trajectories provided with the task models. LeWM can be trained on both pseudo-expert and exploratory behavior-policy data [Maes et al., 2026]. Both memory settings provide ordered observations, temporal indices, and episode boundaries. Direct and the ranking controllers can also use the recorded action values.

The main study uses 128 reserved episodes per task, with one standard start and two preassigned five-action prefixes. Each prefix is a recorded five-action block from the task’s training trajectories. Before control, we clip its actions to the action bounds and execute them while keeping the goal fixed. We sample queries and assign prefixes with fixed random seeds, so every controller sees the same queries, goals, and prefixes. Queries come from the released validation split for Cube, Reacher, and TwoRoom and from the released training split for PushT. Queries use previously unused eligible episodes. For every task, all 128 query episodes are excluded from the retrieval memory and from the trajectories that supply perturbation prefixes. The remaining memories contain 8,000, 16,688, 8,000, and 8,000 episodes for Cube, PushT, Reacher, and TwoRoom. We build the retrieval indices and horizon-specific statistics from these memories with the same frozen features and Equation (5).

All tasks reuse their supplied pretrained checkpoints. Retrieval memory, query episodes, and perturbation prefixes are fixed before evaluation and shared by all controllers.

The goal offset  $H$  and allowance  $B$  depend on the task (Table 5). The perturbation prefix is executed before the policy’s action allowance begins. Policy time then starts at zero and increments by the number of executed primitives, so retrieval uses  $\max(5, H - t)$ . The environment clock continues through the prefix. If an episode succeeds or terminates before policy entry, we assign the same outcome to every controller. This includes success at the initial state or during a prefix. Success rates include these outcomes for all assigned queries.

### A.2 Physical success

Evaluation uses the following task-specific physical predicates. They are distinct from the latent distances used for action selection.

- **Cube:** the selected object’s three-dimensional position is at most 0.04 m from its target. The single-cube data-collection environment hides the target marker. Neither orientation nor gripper position is part of this predicate.
- **PushT:** the Euclidean norm over the concatenated agent and object position errors is strictly below 20 environment units, and the shortest circular object-angle error is below  $\pi/9$ .
- **Reacher:** each of the two joint-position errors has absolute value below 0.05 rad. The errors use direct joint-angle differences.
- **TwoRoom:** the Euclidean distance from the two-dimensional agent position to its goal is below 16 environment units.

We check success initially and after every primitive, using the task predicate at the stopping state to determine the outcome. A controller stops on success, on terminal failure, or when its primitive-action allowance runs out, so execution can end before a full block. The matched construction

restores the same physical state and goal before running each controller. We use simulator coordinates to restore matched starts and to judge physical success, whereas the controller scores encoded images.

### A.3 Shared model and search implementation

The pretrained model belongs to the joint-embedding family [Assran et al., 2023, Balestriero and LeCun, 2025, Maes et al., 2026]. All controllers use the same task-specific frozen encoder. The predictive controllers also share the frozen forward model. Images are resized to 224 pixels with ImageNet normalization, and predicted endpoints are compared in the 192-dimensional projected latent space. Each prediction starts from the current image alone.

We keep LeWM’s pretrained action normalizer unchanged and compute retrieval statistics from the active memory. For a raw action block  $u$ ,  $F(z, u)$  applies this normalization before prediction. The recorded-action controllers and CEM clip proposals to the action bounds before scoring and execute exactly the clipped actions. Clipping leaves the target unchanged: it remains the encoding of the recorded successor image.

Unlike LeWM’s multiblock planner, our three CEM variants search over a single five-action block. They use 30 iterations, 300 candidates, 30 elites, an initial normalized mean of zero, initial standard deviation  $1/3$ , and an elite-standard-deviation floor of  $10^{-5}$ . The search maintains means and standard deviations in normalized coordinates and includes the current mean as the first candidate in each population. For scoring, we clip candidates to the action bounds expressed in normalized coordinates. The elite mean and standard deviation are computed from the unclipped proposals. After selection, we map the chosen block back to raw actions and clip it to the action bounds to remove round-off at the boundary. Finally, we compare the clipped final mean with the clipped initial mean and keep the initial mean unless the final mean has lower cost. For a given task and query, all three target variants and all start conditions use the same random seed. Each search starts from the fixed Gaussian prior. After the 30 populations, the final and initial means are scored together, giving 31 batches and 9,002 predicted blocks per decision.

Direct and the three ranking controllers use the same retrieval rule, and its eight starts can come from the same episode. AP-rank and final-goal rank share the first candidate predictions exactly, and the CEM controllers share the first prior and search noise. Once the policies choose different actions, their later states, retrieved records, and random draws can differ.

### A.4 Aggregation and prediction work

To aggregate perturbed success rates, we first average the two starts of each query and then average queries within a task. All four tasks receive equal weight in task means. Throughout this appendix, we report standard starts and the perturbation average. Original-offset results are reused across comparisons.

To measure prediction work, we count the five-action blocks predicted until the episode ends. Ranking evaluates eight blocks in one batch per decision, and single-block CEM evaluates  $300I + 2$  blocks for  $I$  iterations. LeWM evaluates 45,000 blocks per decision using its multiblock search and executes up to 25 primitive actions before replanning. Direct uses no transition prediction for action selection. Encoding, target construction, retrieval, and interaction also contribute to elapsed runtime.

### A.5 Observation-only deployment

Observation-only memory supplies ordered images, their frozen features, episode boundaries, and temporal indices. These identify the retrieved segment and the observed target. CEM generates candidate actions and passes them to the frozen predictor, which was pretrained with actions. Both memory settings use the same retrieval and target construction.

### A.6 Algorithm and implementation details

Algorithm 1 gives the AP-rank procedure used in every main-study task and condition. Section 3 defines how it retrieves a record, sets a target, selects a clipped action block, and replans after execution. The preceding protocol defines the paired starts and physical success criteria.

LeWM keeps its released action mapping and execution settings (Appendix D.2), and the independent supporting study uses the mapping described in Appendix F.

---

**Algorithm 1** AP-rank with an observed target for a frozen predictor

---

**Require:** frozen  $E, F$ ; memory  $\{\mathcal{B}_h\}$ ; current image  $o_0$ ; goal image  $o_g$ ; supplied  $H$ ; action allowance  $B$

- 1:  $L \leftarrow 5$ ;  $k \leftarrow 8$ ;  $z_g \leftarrow E(o_g)$ ;  $t \leftarrow 0$
- 2: **while**  $t < B$ , the goal is not reached, and the environment is active **do**
- 3:    $z_t \leftarrow E(o_t)$ ;  $h \leftarrow \max\{L, H - t\}$
- 4:   retrieve  $(s_1, \dots, s_k)$  from  $\mathcal{B}_h$  using Equation (5)
- 5:    $q_{\text{obs}} \leftarrow E(o_{s_1+L})$ ;  $\mathcal{C} \leftarrow \{P(u_{s_1}), \dots, P(u_{s_k})\}$
- 6:   predict  $\{F(z_t, u) : u \in \mathcal{C}\}$  in one batch
- 7:   select  $\hat{u}$  by Equation (7), breaking ties by retrieval order
- 8:   execute up to  $L$  primitives from  $\hat{u}$ , stopping at success, termination, or  $B$
- 9:   increase  $t$  by the executed primitives and observe the next image  $o_t$
- 10: **end while**

---

The current observation is encoded with the same frozen encoder as the memory before retrieval and scoring. For each horizon  $h$ , the retrieval-key statistics are computed from the records in  $\mathcal{B}_h$ . Coordinate-wise population standard deviations are floored at  $10^{-4}$ , and the source, distant endpoint, and displacement parts have equal weights.  $\mathcal{B}_h$  and these statistics stay fixed during evaluation. Unlike our controllers, LeWM’s planner accumulates up to three context blocks within its rollout.

All experiments run on NVIDIA H20 GPUs with PyTorch. The tasks use their released MuJoCo-based simulators.

## B Properties of Target-Based Selection

### B.1 A continuous counterexample with smooth dynamics

*Proof of Proposition 1.* Let  $\mathcal{X} = \{c(s) : s \in \mathbb{R}\}$ , where  $c(s) = (s, s^2)$ , and use the identity encoder. A primitive action  $a \in [0, 1/8]$  has transition

$$T((x, y), a) = (x + a, y + 2xa + a^2).$$

This polynomial map is the unit-time flow of  $\dot{x} = a$ ,  $\dot{y} = 2xa$  under constant control and preserves  $\mathcal{X}$ . Thus a five-action block  $u \in \mathcal{C} = [0, 1/8]^5$  has exact endpoint

$$F(c(s), u) = c\left(s + \sum_{i=1}^5 a_i\right).$$

Both controllers globally minimize squared endpoint distance over the entire continuous set  $\mathcal{C}$ , execute the chosen block, and observe again. Only the target differs.

For the final goal  $g = c(2)$ , write

$$D(s) = \|c(s) - g\|^2 = (s - 2)^2 + (s^2 - 4)^2, \quad D'(s) = 2(s - 2)(2s^2 + 4s + 1).$$

The quadratic roots are  $-1 \pm 1/\sqrt{2}$ , so  $D'(s) > 0$  throughout  $[-3/2, -3/8]$ . From any  $s_0 \in [-3/2, -1]$ , an admissible block ends at  $c(s_0 + d_u)$  with  $0 \leq d_u \leq 5/8$ , within that increasing region. Every  $d_u > 0$  increases final-goal distance. Nonnegative actions make the zero block the unique global minimizer. It leaves the state unchanged, so final-goal selection never reaches  $g$ .

Now take one feasible recorded trajectory with consecutive block endpoints

$$(r_0, r_1, r_2, r_3, r_4, r_5, r_6) = \left(-\frac{3}{2}, -\frac{7}{8}, -\frac{1}{4}, \frac{3}{8}, 1, \frac{13}{8}, 2\right).$$

Its first five blocks use  $a_i = 1/8$  and the last uses  $a_i = 3/40$ . Supply the observed targets  $c(r_1), \dots, c(r_6)$  in order. For every allowed  $s_0$ , the first increment  $r_1 - s_0 \in [1/8, 5/8]$  is feasible. Each later increment is also positive and at most  $5/8$ . At each decision a block attains zero target cost, so every global minimizer reaches that target exactly. Induction gives the final goal in six blocks. Ties between primitive sequences do not change their endpoint, and primitive-level stopping can only shorten the final block.  $\square$

The dynamics admit a continuous route from every stated start. Failure under final-goal scoring comes from rejecting the temporary increase in distance needed to follow it. The prescribed recorded sequence isolates the target’s role: with the same exact dynamics and the same one-block global minimization, it yields a goal-reaching controller.

**The failure persists with symmetric actions.** Allow  $a_i \in [-1/8, 1/8]$  and write  $\alpha = -1 - 1/\sqrt{2}$ ,  $\beta = -1 + 1/\sqrt{2}$ . Below  $\beta$ ,  $D$  decreases to  $\alpha$  and then increases. From  $s_0 \in [-3/2, -1]$ , the reachable interval  $[s_0 - 5/8, s_0 + 5/8]$  lies below  $\beta$ . Its minimizing coordinate is  $\max\{\alpha, s_0 - 5/8\}$ , and repeating this rule reaches  $\alpha$  within two blocks. Since  $\alpha + 5/8 < \beta$ , subsequent minimizing blocks also end at  $\alpha$ . The same observed-target sequence remains feasible from each original start and reaches the goal. Allowing both action directions thus preserves failure under final-goal scoring.

## B.2 How endpoint errors affect action selection

At a fixed live state and target  $q$ , let  $e_u$  be an action block’s exact encoded endpoint and  $r_u = F(z_t, u) - e_u$  its prediction error. The resulting error in target cost is

$$\xi_u(q) = J_q(u) - \|e_u - q\|^2 = 2(e_u - q)^\top r_u + \|r_u\|^2.$$

The predicted cost difference between  $u$  and  $v$  therefore differs from the exact one by  $\xi_u(q) - \xi_v(q)$ . Two endpoint errors with the same size can change action costs differently: the dot product depends on the direction of each error relative to the target. If both costs change by the same amount, their ordering stays unchanged.

If exact target costs favor  $u$  over  $v$  by a gap  $m > 0$ , that ordering is preserved whenever  $\xi_u(q) - \xi_v(q) < m$ . This explains why endpoint error and selection regret provide complementary information in Table 4: the first measures endpoint accuracy, while the second measures the cost of the action chosen under the target.

Proposition 1 concerns the target itself. With  $r_u = 0$ , every predicted target cost is exact, yet final-goal scoring still favors staying still in the construction. Changing the target changes those costs without changing the dynamics.

## C RQ1: Target Choice and Control

The main target comparisons use the shared protocol in Appendix A. This section gives target-model selection and the physical behavior behind the success rates in Table 1.

### C.1 Learned intermediate targets

We train a target predictor on the same memory episodes available to Anchored Planning. For each task, a fixed episode split reserves five percent of memory for validation. Control-query episodes are excluded from both splits. To form a training example, we sample an offset  $h$  uniformly from  $5, \dots, H_{\text{task}}$ , where  $H_{\text{task}}$  is the task’s main goal offset, and then choose a valid source position  $s$  uniformly among the training positions at that offset. The input is  $(z_s, z_{s+h}, z_{s+h} - z_s, h/H_{\text{task}})$ , and the target is the observed successor  $z_{s+5}$ . All encodings come from the frozen LeWM encoder.

The predictor is an MLP with three hidden layers, each using LayerNorm and GELU. We compare widths 512 and 1024, with either an absolute output or a residual added to  $z_s$ . Inputs and outputs are standardized coordinate-wise using 262,144 sampled training transitions. We train with AdamW using batch size 1024, learning rate  $3 \times 10^{-4}$ , weight decay  $10^{-4}$ , and cosine decay over at most 50,000 updates. Validation uses the same 4096 held-out transitions for all four configurations. We evaluate every 500 updates and stop after ten checks without improvement, once at least 5000 updates have run. The selected checkpoint minimizes unstandardized latent-coordinate MSE, which gives absolute and residual models the same selection scale.

At execution, the MLP receives the current encoding, final-goal encoding, their difference, and  $h(t)/H_{\text{task}}$ . Its output replaces the scoring target for either CEM or ranking. CEM keeps its initialization and search settings, ranking scores the same retrieved blocks, and both keep the frozen predictor, action clipping, and execution rule of the main study.

Table 2 relates target accuracy to control. To measure target accuracy, we use 4096 held-out transitions per task and exclude each source episode from retrieval. Successor errors and nearest-memory distances are squared latent distances. An observed target has zero distance to memory by construction.

Table 6: **Selecting learned targets on held-out memory.** We average squared error over latent coordinates to compute validation MSE. Bold marks the selected configuration.

| Task    | Width | Output   | Best step | Validation MSE  |
|---------|-------|----------|-----------|-----------------|
| Cube    | 512   | Absolute | 48,500    | 0.104926        |
|         | 512   | Residual | 48,500    | 0.104579        |
|         | 1024  | Absolute | 41,000    | 0.104422        |
|         | 1024  | Residual | 48,500    | <b>0.102646</b> |
| PushT   | 512   | Absolute | 48,000    | 0.023950        |
|         | 512   | Residual | 48,000    | 0.024007        |
|         | 1024  | Absolute | 47,500    | 0.020210        |
|         | 1024  | Residual | 45,500    | <b>0.019658</b> |
| Reacher | 512   | Absolute | 11,500    | 0.391840        |
|         | 512   | Residual | 21,500    | <b>0.389188</b> |
|         | 1024  | Absolute | 11,500    | 0.391255        |
|         | 1024  | Residual | 21,500    | 0.389396        |
| TwoRoom | 512   | Absolute | 14,500    | <b>0.697172</b> |
|         | 512   | Residual | 15,500    | 0.702978        |
|         | 1024  | Absolute | 8,000     | 0.697239        |
|         | 1024  | Residual | 7,500     | 0.704444        |

## C.2 Physical behavior along a rollout

Physical error is measured in units of the success thresholds. For Cube and TwoRoom,  $\hat{e}_t$  is position error divided by 0.04 m and 16 units, respectively. For Reacher, it is the larger joint error divided by 0.05 rad. For PushT, it is the maximum of combined position error divided by 20 and the wrapped object-angle error divided by  $\pi/9$ . We apply the success criteria defined in Appendix A.

A failed rollout stalls when its task-relevant coordinates remain within one quarter of their success thresholds from the state at the beginning of the last  $W = 10$  decisions. Shorter rollouts use their available window. A successful rollout makes a detour when  $\hat{e}_t$  rises by at least  $\delta = 0.5$  above its previous running minimum.

Table 7 gives the rates with their eligible counts, and Table 8 varies the window and threshold for the three CEM target rules.

## C.3 A matched PushT rollout

We examine main-study queries in a fixed order, testing the first prefix before the second for each query. Figure 6 shows the physical states throughout the first rollout in which AP-rank succeeds and Direct fails.

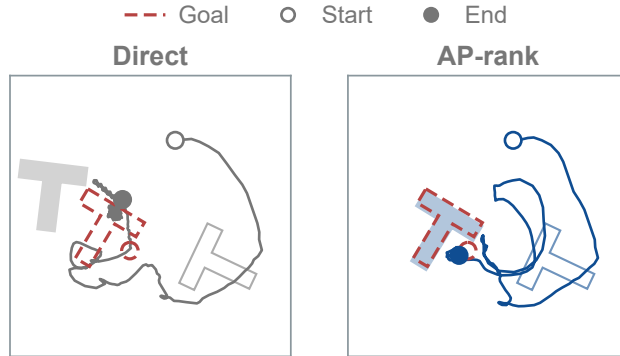


Figure 6: **Predictive ranking changes a matched PushT execution.** Both controllers share the perturbed start and goal. Curves trace the agent’s path. Circles show the agent, and T shapes show the pushed block. Hollow and filled shapes show initial and terminal poses, and dashed outlines mark the goal.

Table 7: **Physical behavior at the original goal offsets.** Each percentage is followed by the number of positive cases and eligible rollouts. With  $W = 10$ , we measure stalling over failed rollouts in which the policy made at least one decision. With  $\delta = 0.5$ , we measure detours over successful rollouts that entered policy control. For perturbed behavior, we pool eligible rollouts from both prefixes.

| Task    | Controller          | Standard        |                 | Perturbed       |                 |
|---------|---------------------|-----------------|-----------------|-----------------|-----------------|
|         |                     | Stalling        | Detour          | Stalling        | Detour          |
| Cube    | LeWM                | 39.8 (43/108)   | 10.5 (2/19)     | 48.2 (108/224)  | 6.7 (2/30)      |
|         | CEM (final goal)    | 100.0 (119/119) | 37.5 (3/8)      | 100.0 (249/249) | 40.0 (2/5)      |
|         | CEM (learned)       | 79.2 (38/48)    | 60.8 (48/79)    | 89.2 (140/157)  | 68.0 (66/97)    |
|         | AP-CEM              | 86.8 (59/68)    | 66.1 (39/59)    | 92.2 (166/180)  | 73.0 (54/74)    |
|         | CEM (transported)   | 91.3 (73/80)    | 68.1 (32/47)    | 95.4 (186/195)  | 71.2 (42/59)    |
|         | Direct              | 87.0 (20/23)    | 60.6 (63/104)   | 94.0 (126/134)  | 64.2 (77/120)   |
|         | Final-goal rank     | 93.8 (45/48)    | 55.7 (44/79)    | 99.3 (148/149)  | 61.0 (64/105)   |
|         | Learned-target rank | 100.0 (11/11)   | 64.7 (75/116)   | 89.1 (114/128)  | 63.5 (80/126)   |
|         | AP-rank             | 100.0 (34/34)   | 61.3 (57/93)    | 92.8 (129/139)  | 67.0 (77/115)   |
| PushT   | LeWM                | 28.6 (34/119)   | 88.9 (8/9)      | 28.0 (65/232)   | 83.3 (20/24)    |
|         | CEM (final goal)    | 80.0 (100/125)  | 33.3 (1/3)      | 78.7 (196/249)  | 42.9 (3/7)      |
|         | CEM (learned)       | 80.3 (49/61)    | 98.5 (66/67)    | 81.7 (107/131)  | 99.2 (124/125)  |
|         | AP-CEM              | 92.3 (36/39)    | 100.0 (89/89)   | 83.3 (75/90)    | 100.0 (166/166) |
|         | CEM (transported)   | 87.8 (43/49)    | 100.0 (79/79)   | 78.2 (111/142)  | 100.0 (114/114) |
|         | Direct              | 95.2 (59/62)    | 100.0 (66/66)   | 91.0 (191/210)  | 100.0 (46/46)   |
|         | Final-goal rank     | 90.5 (86/95)    | 100.0 (33/33)   | 88.3 (197/223)  | 100.0 (33/33)   |
|         | Learned-target rank | 93.1 (54/58)    | 100.0 (70/70)   | 87.3 (144/165)  | 98.9 (90/91)    |
|         | AP-rank             | 90.7 (39/43)    | 100.0 (85/85)   | 89.6 (120/134)  | 100.0 (122/122) |
| Reacher | LeWM                | 0.0 (0/43)      | 87.1 (74/85)    | 0.0 (0/80)      | 90.8 (157/173)  |
|         | CEM (final goal)    | 0.0 (0/95)      | 51.5 (17/33)    | 0.0 (0/198)     | 47.3 (26/55)    |
|         | CEM (learned)       | 0.0 (0/58)      | 92.9 (65/70)    | 0.0 (0/105)     | 93.9 (139/148)  |
|         | AP-CEM              | 0.0 (0/31)      | 96.9 (94/97)    | 0.0 (0/54)      | 97.5 (194/199)  |
|         | CEM (transported)   | 0.0 (0/24)      | 96.2 (100/104)  | 0.0 (0/50)      | 97.5 (198/203)  |
|         | Direct              | 0.0 (0/1)       | 97.6 (124/127)  | 0.0 (0/8)       | 99.2 (243/245)  |
|         | Final-goal rank     | 0.0 (0/12)      | 94.0 (109/116)  | 0.0 (0/32)      | 95.0 (210/221)  |
|         | Learned-target rank | 0.0 (0/6)       | 100.0 (122/122) | 0.0 (0/8)       | 99.6 (244/245)  |
|         | AP-rank             | 0.0 (0/4)       | 98.4 (122/124)  | 0.0 (0/11)      | 99.6 (241/242)  |
| TwoRoom | LeWM                | 0.0 (0/106)     | 68.2 (15/22)    | 0.0 (0/213)     | 51.2 (22/43)    |
|         | CEM (final goal)    | 33.3 (42/126)   | 0.0 (0/2)       | 32.1 (81/252)   | 0.0 (0/4)       |
|         | CEM (learned)       | 0.0 (0/60)      | 27.9 (19/68)    | 0.0 (0/115)     | 27.0 (38/141)   |
|         | AP-CEM              | 19.1 (13/68)    | 73.3 (44/60)    | 15.4 (21/136)   | 75.8 (91/120)   |
|         | CEM (transported)   | 41.7 (35/84)    | 59.1 (26/44)    | 39.8 (66/166)   | 64.4 (58/90)    |
|         | Direct              | 0.0 (0/1)       | 96.9 (123/127)  | – (0/0)         | 99.2 (254/256)  |
|         | Final-goal rank     | 7.1 (1/14)      | 93.9 (107/114)  | 6.9 (2/29)      | 96.5 (219/227)  |
|         | Learned-target rank | 25.0 (1/4)      | 97.6 (121/124)  | 0.0 (0/4)       | 96.8 (244/252)  |
|         | AP-rank             | 0.0 (0/2)       | 98.4 (124/126)  | – (0/0)         | 99.2 (254/256)  |

Table 8: **Behavior across thresholds.** We average rates (%) equally across the four tasks, computing each rate from that task’s eligible rollout counts.

|  | Controller       | Start     | Stalling window |      |      | Detour threshold |      |      |
|--|------------------|-----------|-----------------|------|------|------------------|------|------|
|  |                  |           | $W = 5$         | 10   | 20   | $\delta = 0.25$  | 0.5  | 1.0  |
|  | CEM (final goal) | Standard  | 55.1            | 53.3 | 49.9 | 68.6             | 30.6 | 29.1 |
|  | CEM (final goal) | Perturbed | 54.9            | 52.7 | 49.7 | 46.4             | 32.5 | 19.0 |
|  | CEM (learned)    | Standard  | 46.1            | 39.9 | 31.4 | 77.6             | 70.0 | 53.7 |
|  | CEM (learned)    | Perturbed | 45.6            | 42.7 | 32.1 | 79.3             | 72.0 | 58.9 |
|  | AP-CEM           | Standard  | 51.7            | 49.5 | 40.5 | 90.6             | 84.1 | 70.7 |
|  | AP-CEM           | Perturbed | 52.2            | 47.7 | 39.3 | 91.5             | 86.6 | 71.8 |

## D RQ2: Search Budget and Goal Distance

### D.1 Search budgets

In the budget sweep, we vary CEM iterations over  $\{1, 2, 5, 10, 30\}$  while preserving the main queries, perturbation prefixes, frozen models, and action clipping. At every budget, we use the same query seed and initial proposals, then draw candidates for the specified number of iterations. The largest-budget points reuse the main outcomes. Table 9 gives both start conditions.

Table 9: **Success across search budgets.** We report success (%) on the main queries at the original offsets. Only the number of CEM iterations changes, and each decision evaluates  $300I + 2$  predicted blocks. For perturbed starts, we average the two starts of each query.

| Task    | Iterations | Final goal |           | Learned target |           | Observed target |           |
|---------|------------|------------|-----------|----------------|-----------|-----------------|-----------|
|         |            | Standard   | Perturbed | Standard       | Perturbed | Standard        | Perturbed |
| Cube    | 1          | 3.9        | 2.0       | 11.7           | 6.6       | 10.9            | 7.0       |
|         | 2          | 3.1        | 2.3       | 40.6           | 26.2      | 26.6            | 15.6      |
|         | 5          | 3.1        | 2.3       | 52.3           | 34.0      | 36.7            | 26.6      |
|         | 10         | 4.7        | 3.9       | 59.4           | 35.2      | 44.5            | 29.7      |
|         | 30         | 7.0        | 2.7       | 62.5           | 38.7      | 46.9            | 29.7      |
| PushT   | 1          | 1.6        | 2.0       | 3.9            | 3.9       | 3.1             | 1.6       |
|         | 2          | 1.6        | 2.0       | 25.8           | 27.0      | 8.6             | 10.9      |
|         | 5          | 1.6        | 2.0       | 44.5           | 41.8      | 43.8            | 43.4      |
|         | 10         | 1.6        | 2.7       | 48.4           | 44.5      | 69.5            | 68.0      |
|         | 30         | 2.3        | 2.7       | 52.3           | 48.8      | 69.5            | 64.8      |
| Reacher | 1          | 16.4       | 15.2      | 48.4           | 50.0      | 59.4            | 56.6      |
|         | 2          | 18.8       | 18.4      | 50.8           | 53.9      | 64.8            | 67.2      |
|         | 5          | 24.2       | 20.3      | 60.2           | 59.8      | 78.9            | 75.8      |
|         | 10         | 23.4       | 23.4      | 63.3           | 60.5      | 85.9            | 75.4      |
|         | 30         | 25.8       | 22.7      | 54.7           | 59.0      | 75.8            | 78.9      |
| TwoRoom | 1          | 0.0        | 0.0       | 34.4           | 35.2      | 7.0             | 9.0       |
|         | 2          | 0.0        | 0.0       | 39.1           | 41.0      | 19.5            | 20.3      |
|         | 5          | 0.8        | 0.0       | 46.9           | 49.2      | 37.5            | 37.1      |
|         | 10         | 0.8        | 1.6       | 55.5           | 53.5      | 43.0            | 45.3      |
|         | 30         | 1.6        | 1.6       | 53.1           | 55.1      | 46.9            | 46.9      |

### D.2 LeWM configuration

LeWM uses the planner released with the model in the `stable-worldmodel` library (version 0.0.6). Its five model blocks each contain five primitive actions. Before replanning, the policy executes all five blocks unless success, termination, or the action allowance stops it earlier. CEM uses 300 candidates, 30 iterations, 30 elites, initial standard deviation 1, and the released history and mean updates. Its actions are de-normalized with the pretrained normalizer and sent directly to the environment. Starts, goals, action allowances, checkpoints, and success predicates match the target comparisons. Our CEM variants, by contrast, share one search setting and differ only in the target.

**Released evaluation protocol.** The released LeWM evaluator samples valid start states from the supplied datasets, takes the goal image 24 actions after the start, and allows 50 actions. Our  $H = 25$  queries place the goal 25 actions after the start and allow 25 actions on Cube and 50 on the other tasks. The two protocols also start episodes differently. The released evaluator scores the dataset image of the start and, on Cube, restores the full recorded simulator state, including joint positions and velocities. We instead render the restored start, and every controller in our study begins from a state reconstructed from the public observation. Finally, the released configuration uses 30 CEM iterations on every task, although Maes et al. [2026] report ten outside PushT.

Table 10 lists the published rates next to runs of the released evaluator and of our implementation with the same queries, action mapping, and CEM random seeds. The two implementations give the same outcome on 599 of 600 episodes.

Table 10: **LeWM under its released evaluation protocol.** We compare published success (%) from Figure 6 of Maes et al. [2026] with measurements from the released evaluator and our implementation on the same queries. Measurements are averaged over three seeds with 50 queries each, including the seed specified in the released configuration.

| Task    | Paper report | Released evaluator | Our implementation |
|---------|--------------|--------------------|--------------------|
| Cube    | 74.0         | 74.7               | 74.7               |
| PushT   | 96.0         | 92.0               | 91.3               |
| Reacher | 86.0         | 80.0               | 80.0               |
| TwoRoom | 87.0         | 87.3               | 87.3               |

On Cube and PushT, query sampling produces the largest change among the factors we varied one at a time: with the released sampling rule, LeWM success in our  $H = 25$  setting rises from 32.0% to 71.9% on Cube and from 60.9% to 82.8% on PushT, whereas a 50-action allowance alone raises Cube to 36.7%. Applying the released success criteria to 512 main-study episodes changes no outcome.

**Five-action replanning.** The additional variant replans after at most five primitive actions instead of 25. It keeps the released lookahead, candidate count, iterations, elites, initial deviation, and warm start, and uses the main queries, goals, and action allowances (Table 11).

Table 11: **LeWM with five-action replanning.** The variant keeps the 25-action lookahead and the released CEM settings and changes only execution, which now covers one five-action block. Evaluation uses the main queries and original goal offsets, with 128 standard starts and 256 perturbed starts per task. Work is the mean number of predicted five-action blocks per episode, in thousands as in Table 13.

| Task    | Success (%) |           | Work ( $10^3$ blocks) |           |
|---------|-------------|-----------|-----------------------|-----------|
|         | Standard    | Perturbed | Standard              | Perturbed |
| Cube    | 13.3        | 13.3      | 1,189                 | 1,197     |
| PushT   | 7.0         | 8.2       | 2,402                 | 2,374     |
| Reacher | 73.4        | 77.3      | 1,388                 | 1,322     |
| TwoRoom | 27.3        | 25.8      | 1,500                 | 1,509     |

### D.3 Recorded goal offsets

Each goal is the observation at the selected offset from the original query source. We add  $H = 25$ , 50, and 100 for Cube, PushT, and Reacher, and  $H = 25$  and 50 for TwoRoom. We set the action allowance to preserve each task’s original  $B/H$ . The learned target retains its training normalization  $H_{\text{task}}$ . Original-offset points reuse the main outcomes. Table 12 gives synthesis and ranking results.

### D.4 Prediction work

Table 13 reports prediction work at the original offsets, counted as in Appendix A.4. LeWM can execute up to 25 actions between decisions, whereas the other predictive controllers execute up to five.

Table 13: **Mean episode prediction work at the original offsets.** Values are thousands of predicted five-action blocks per episode. Columns marked Std. show standard starts, and those marked Pert. show the average of each query’s perturbed starts.

| Controller          | Cube   |        | PushT  |        | Reacher |        | TwoRoom |        |
|---------------------|--------|--------|--------|--------|---------|--------|---------|--------|
|                     | Std.   | Pert.  | Std.   | Pert.  | Std.    | Pert.  | Std.    | Pert.  |
| LeWM                | 235.20 | 241.88 | 514.69 | 505.37 | 255.59  | 244.51 | 320.98  | 321.50 |
| CEM (final goal)    | 252.62 | 263.17 | 493.56 | 491.10 | 407.76  | 425.31 | 354.95  | 354.95 |
| CEM (learned)       | 142.27 | 191.54 | 375.06 | 387.40 | 393.91  | 379.98 | 250.65  | 247.94 |
| AP-CEM              | 170.26 | 206.84 | 326.11 | 338.63 | 323.58  | 313.17 | 262.61  | 265.07 |
| CEM (transported)   | 193.40 | 223.26 | 347.84 | 393.80 | 306.56  | 310.50 | 289.19  | 291.37 |
| Direct              | 0.00   | 0.00   | 0.00   | 0.00   | 0.00    | 0.00   | 0.00    | 0.00   |
| Final-goal rank     | 0.12   | 0.16   | 0.38   | 0.42   | 0.20    | 0.20   | 0.15    | 0.16   |
| Learned-target rank | 0.08   | 0.15   | 0.33   | 0.37   | 0.24    | 0.23   | 0.14    | 0.14   |
| AP-rank             | 0.10   | 0.15   | 0.30   | 0.34   | 0.24    | 0.24   | 0.14    | 0.14   |

Table 12: **Success across recorded goal offsets.** To compare success (%) across offsets, we keep query sources fixed and move the goal to the observation  $H$  actions later, preserving each task’s  $B/H$ . Prediction lookahead and target-model training remain unchanged. Results are shown for standard starts (Std.) and for the average of each query’s perturbed starts (Pert.).

| <i>Action synthesis</i> |     |      |       |            |       |         |       |        |       |
|-------------------------|-----|------|-------|------------|-------|---------|-------|--------|-------|
| Task                    | $H$ | LeWM |       | Final goal |       | Learned |       | AP-CEM |       |
|                         |     | Std. | Pert. | Std.       | Pert. | Std.    | Pert. | Std.   | Pert. |
| Cube                    | 25  | 32.0 | 33.6  | 28.9       | 26.2  | 75.0    | 53.9  | 63.3   | 49.6  |
|                         | 50  | 16.4 | 12.1  | 7.0        | 4.3   | 58.6    | 37.9  | 47.7   | 30.5  |
|                         | 100 | 18.8 | 13.3  | 8.6        | 9.0   | 55.5    | 34.8  | 32.0   | 25.4  |
|                         | 150 | 15.6 | 12.5  | 7.0        | 2.7   | 62.5    | 38.7  | 46.9   | 29.7  |
| PushT                   | 25  | 60.9 | 57.8  | 51.6       | 46.9  | 91.4    | 75.4  | 91.4   | 87.5  |
|                         | 50  | 22.7 | 21.9  | 6.3        | 9.0   | 76.6    | 63.7  | 82.0   | 81.6  |
|                         | 100 | 10.9 | 9.4   | 3.9        | 4.3   | 57.0    | 48.4  | 77.3   | 73.8  |
|                         | 140 | 7.0  | 9.4   | 2.3        | 2.7   | 52.3    | 48.8  | 69.5   | 64.8  |
| Reacher                 | 25  | 68.8 | 67.6  | 66.4       | 66.0  | 57.8    | 57.8  | 67.2   | 71.1  |
|                         | 50  | 95.3 | 92.6  | 57.0       | 52.3  | 53.1    | 53.5  | 81.3   | 76.2  |
|                         | 100 | 83.6 | 82.0  | 35.2       | 35.2  | 48.4    | 52.3  | 84.4   | 76.6  |
|                         | 150 | 66.4 | 68.8  | 25.8       | 22.7  | 54.7    | 59.0  | 75.8   | 78.9  |
| TwoRoom                 | 25  | 77.3 | 74.2  | 30.5       | 34.8  | 99.2    | 99.2  | 89.1   | 85.9  |
|                         | 50  | 36.7 | 35.5  | 5.5        | 6.6   | 94.5    | 94.1  | 72.7   | 73.8  |
|                         | 100 | 17.2 | 16.8  | 1.6        | 1.6   | 53.1    | 55.1  | 46.9   | 46.9  |

| <i>Recorded-action selection</i> |     |            |       |         |       |         |       |
|----------------------------------|-----|------------|-------|---------|-------|---------|-------|
| Task                             | $H$ | Final goal |       | Learned |       | AP-rank |       |
|                                  |     | Std.       | Pert. | Std.    | Pert. | Std.    | Pert. |
| Cube                             | 25  | 92.2       | 60.9  | 88.3    | 57.8  | 92.2    | 59.0  |
|                                  | 50  | 68.0       | 43.8  | 83.6    | 48.4  | 70.3    | 44.1  |
|                                  | 100 | 59.4       | 34.8  | 78.9    | 43.8  | 64.8    | 37.1  |
|                                  | 150 | 62.5       | 41.8  | 91.4    | 50.0  | 73.4    | 45.7  |
| PushT                            | 25  | 89.8       | 68.0  | 96.1    | 68.4  | 95.3    | 71.1  |
|                                  | 50  | 67.2       | 36.7  | 81.3    | 48.4  | 84.4    | 55.9  |
|                                  | 100 | 26.6       | 14.1  | 64.1    | 38.7  | 71.1    | 46.9  |
|                                  | 140 | 25.8       | 12.9  | 54.7    | 35.5  | 66.4    | 47.7  |
| Reacher                          | 25  | 92.2       | 93.0  | 87.5    | 88.3  | 91.4    | 91.8  |
|                                  | 50  | 97.7       | 97.7  | 94.5    | 93.8  | 97.7    | 96.5  |
|                                  | 100 | 95.3       | 94.9  | 97.7    | 96.9  | 97.7    | 96.1  |
|                                  | 150 | 90.6       | 87.5  | 95.3    | 96.9  | 96.9    | 95.7  |
| TwoRoom                          | 25  | 100.0      | 99.2  | 100.0   | 99.6  | 100.0   | 100.0 |
|                                  | 50  | 94.5       | 94.1  | 100.0   | 99.6  | 100.0   | 100.0 |
|                                  | 100 | 89.1       | 88.7  | 96.9    | 98.4  | 98.4    | 100.0 |

## E RQ3: Memory Targets and Predictive Selection

### E.1 Target offsets, memory size, and retrieval keys

The target-offset ablation uses  $q = z_{s_1+\ell}$  with  $\ell \in \{1, 3, 10\}$  while prediction and execution retain  $L = 5$ . By default, we use  $\ell = L$  and retrieve only sources with a valid successor at the chosen offset. Memory-size ablations use nested episode subsets of 10, 25, and 50 percent, sampled once with a fixed seed, and recompute retrieval standard deviations in each subset. Table 14 gives task-level results for both action rules.

The retrieval-key ablations omit either the distant endpoint or the displacement component. The fixed-span ablation uses  $h = H$  throughout execution. Other retrieval and control settings remain unchanged (Table 15).

Table 14: **Target offsets and memory size.** On the main queries, we measure success (%) while varying the target offset or memory size. By default, the target is the five-step successor and retrieval uses the full memory. Prediction and execution retain five-action blocks throughout.

| Setting        | Cube |       | PushT |       | Reacher |       | TwoRoom |       |
|----------------|------|-------|-------|-------|---------|-------|---------|-------|
|                | Std. | Pert. | Std.  | Pert. | Std.    | Pert. | Std.    | Pert. |
| <i>AP-CEM</i>  |      |       |       |       |         |       |         |       |
| Default        | 46.9 | 29.7  | 69.5  | 64.8  | 75.8    | 78.9  | 46.9    | 46.9  |
| $\ell = 1$     | 43.8 | 31.3  | 34.4  | 29.3  | 50.0    | 44.9  | 43.8    | 46.1  |
| $\ell = 3$     | 61.7 | 42.6  | 68.8  | 60.9  | 91.4    | 90.6  | 57.8    | 56.3  |
| $\ell = 10$    | 19.5 | 15.6  | 58.6  | 57.4  | 87.5    | 89.8  | 50.8    | 48.0  |
| 10% memory     | 39.1 | 25.4  | 59.4  | 58.6  | 75.8    | 76.2  | 52.3    | 48.4  |
| 25% memory     | 39.1 | 27.7  | 64.8  | 63.3  | 79.7    | 76.2  | 50.0    | 46.9  |
| 50% memory     | 43.8 | 30.1  | 67.2  | 61.3  | 76.6    | 76.2  | 46.9    | 47.7  |
| <i>AP-rank</i> |      |       |       |       |         |       |         |       |
| Default        | 73.4 | 45.7  | 66.4  | 47.7  | 96.9    | 95.7  | 98.4    | 100.0 |
| $\ell = 1$     | 71.9 | 44.5  | 30.5  | 21.9  | 93.8    | 94.9  | 82.0    | 84.0  |
| $\ell = 3$     | 71.1 | 44.9  | 49.2  | 35.5  | 95.3    | 96.1  | 98.4    | 97.3  |
| $\ell = 10$    | 71.1 | 44.1  | 59.4  | 34.4  | 96.9    | 95.7  | 99.2    | 99.2  |
| 10% memory     | 63.3 | 42.2  | 53.1  | 39.5  | 94.5    | 94.5  | 99.2    | 99.6  |
| 25% memory     | 67.2 | 44.1  | 58.6  | 43.0  | 96.1    | 95.7  | 98.4    | 98.8  |
| 50% memory     | 73.4 | 45.3  | 61.7  | 43.4  | 93.8    | 95.7  | 99.2    | 99.6  |

Table 15: **Retrieval keys and temporal span.** For AP-rank, we compare success (%) on the main queries. When omitting a key component, we keep the other components unchanged. Fixing  $h = H$  replaces the adaptive span  $\max(5, H - t)$ .

| Setting                  | Cube |       | PushT |       | Reacher |       | TwoRoom |       |
|--------------------------|------|-------|-------|-------|---------|-------|---------|-------|
|                          | Std. | Pert. | Std.  | Pert. | Std.    | Pert. | Std.    | Pert. |
| Full key                 | 73.4 | 45.7  | 66.4  | 47.7  | 96.9    | 95.7  | 98.4    | 100.0 |
| Without distant endpoint | 81.3 | 46.5  | 67.2  | 50.4  | 94.5    | 94.5  | 100.0   | 100.0 |
| Without displacement     | 73.4 | 45.7  | 63.3  | 44.9  | 96.1    | 97.7  | 99.2    | 100.0 |
| Fixed span $h = H$       | 4.7  | 3.1   | 0.0   | 0.0   | 12.5    | 19.1  | 21.1    | 23.0  |

## E.2 Absolute endpoints and transported displacements

The anchoring comparison uses observation-only CEM on the main queries. For a live encoding  $z_t$ , retrieved source encoding  $z_s$ , and successor encoding  $z_{s+L}$ , the two target rules are

$$q_{\text{obs}} = z_{s+L}, \quad q_{\text{transport}} = z_t + z_{s+L} - z_s, \quad L = 5. \quad (9)$$

The live, source, and successor images use the same frozen encoder. Both rules construct targets from ordered observations and use a common Gaussian prior to generate actions. Their difference is  $q_{\text{obs}} - q_{\text{transport}} = z_s - z_t$ : one pursues the recorded endpoint, while the other matches the recorded displacement from the live encoding. The targets coincide when the source and live encodings coincide.

**How the target changes action ordering.** Suppose the current encoding has already moved partway along the recorded displacement. Anchoring requests only the remaining movement to the recorded endpoint. Transport repeats the full displacement from the new state, placing its target beyond that endpoint.

Let  $e = z_t - z_s$  denote the current offset from the recorded source. Applying Equation (2) to the two targets gives

$$\Delta_{q_{\text{obs}}}(u, v) - \Delta_{q_{\text{transport}}}(u, v) = 2(p_u - p_v)^\top e. \quad (10)$$

When the live state moves away from the recorded source, Equation (10) gives the resulting change in the cost difference between the two actions. Its sign determines which action gains relative to the other. Perturbing the start changes the live state on which both rules act. Within each matched start, the target variants share their initial retrieved record.

Table 16: **Anchoring and displacement transport on the main queries.** Under the same CEM search, we compare success (%) when using the observed endpoint or transported motion. At perturbed starts, we average the two starts of each query, and the mean weights all tasks equally.

| Task    | Start     | Observed target | Transported target |
|---------|-----------|-----------------|--------------------|
| Cube    | Standard  | <b>46.9</b>     | 37.5               |
|         | Perturbed | <b>29.7</b>     | 23.8               |
| PushT   | Standard  | <b>69.5</b>     | 61.7               |
|         | Perturbed | <b>64.8</b>     | 44.5               |
| Reacher | Standard  | 75.8            | <b>81.3</b>        |
|         | Perturbed | 78.9            | <b>80.5</b>        |
| TwoRoom | Standard  | <b>46.9</b>     | 34.4               |
|         | Perturbed | <b>46.9</b>     | 35.2               |
| Mean    | Standard  | <b>59.8</b>     | 53.7               |
|         | Perturbed | <b>55.1</b>     | 46.0               |

### E.3 Exact endpoints and the first action block

At each assigned start, the simulator executes the eight retrieved action blocks from the same physical state. The frozen encoder maps their resulting observations to exact candidate endpoints. For endpoint error and selection regret, we include starts for which all eight candidates execute five actions. To obtain endpoint error, we first average over candidates. At perturbed starts, we average the eligible starts of each query and then give queries equal weight.

The first-block intervention compares Direct, the frozen predictor, simulator endpoints, and transported displacement estimates. We score predictor and simulator endpoints against each of the final, observed, and learned targets, whereas displacement uses the observed target. After the selected first block, every rollout uses Direct. When computing success rates, we include all 128 assigned queries per task, including starts resolved before a policy decision.

Table 4 reports endpoint error and regret for standard starts and the perturbation average. Table 17 gives every first-block selector with its common continuation.

Table 17: **Changing only the first action block.** After selecting the first block, we use the same Direct continuation to measure success (%). Predictor and simulator selectors score the same eight candidates against the indicated target. All 128 assigned queries per task are included.

| First-block selector    | Cube |       | PushT |       | Reacher |       | TwoRoom |       |
|-------------------------|------|-------|-------|-------|---------|-------|---------|-------|
|                         | Std. | Pert. | Std.  | Pert. | Std.    | Pert. | Std.    | Pert. |
| Direct                  | 82.0 | 47.7  | 51.6  | 18.0  | 99.2    | 96.9  | 99.2    | 100.0 |
| Predictor / Final       | 67.2 | 43.0  | 39.1  | 18.4  | 96.1    | 97.3  | 100.0   | 100.0 |
| Simulator / Final       | 73.4 | 44.5  | 43.0  | 19.1  | 96.1    | 97.3  | 99.2    | 99.6  |
| Predictor / Learned     | 85.9 | 48.8  | 45.3  | 18.8  | 96.9    | 97.7  | 100.0   | 99.6  |
| Simulator / Learned     | 86.7 | 49.2  | 42.2  | 22.3  | 96.9    | 97.7  | 100.0   | 99.6  |
| Predictor / Observed    | 78.9 | 47.3  | 53.1  | 22.3  | 95.3    | 96.1  | 99.2    | 99.2  |
| Simulator / Observed    | 80.5 | 48.8  | 53.9  | 23.8  | 96.9    | 96.1  | 100.0   | 99.2  |
| Displacement / Observed | 78.1 | 48.4  | 50.8  | 23.0  | 98.4    | 95.7  | 99.2    | 99.6  |

## F Independent-Query Target Replication

The supporting study uses 128 validation episodes per task, disjoint from the main queries, with the same task-specific goal offsets, action allowances, frozen models, and physical success criteria. It passes de-normalized actions directly to the environment, without the clipping to action bounds used in the main study. Each target contrast shares this query set and action mapping.

For CEM, we compare final-goal and observed targets at standard starts, using 30 iterations, 300 candidates, 30 elites, and initial standard deviation  $1/3$ . For ranking, we compare the same targets on eight retrieved action blocks at the standard start and two preassigned five-action perturbations. The goal and memory remain fixed across starts, and we include all assigned outcomes when computing success rates. Table 18 collects the comparisons.

Table 18: **Target effects on independent supporting queries.** For each action rule, we compare success (%) with final-goal and observed targets. Ranking is evaluated at standard and perturbed starts, whereas CEM is evaluated at standard starts. Within each query, we average the two assigned perturbed starts.

| Task    | CEM: Standard |          | Ranking: Standard |          | Ranking: Perturbed |          |
|---------|---------------|----------|-------------------|----------|--------------------|----------|
|         | Final         | Observed | Final             | Observed | Final              | Observed |
| Cube    | 8.6           | 39.1     | 64.8              | 71.1     | 41.4               | 43.4     |
| PushT   | 1.6           | 81.3     | 25.8              | 78.1     | 9.0                | 49.2     |
| Reacher | 25.0          | 75.8     | 92.2              | 95.3     | 88.7               | 95.7     |
| TwoRoom | 0.8           | 39.1     | 85.2              | 100.0    | 88.3               | 99.6     |

Observed targets improve success over final-goal scoring in every task under both action rules. Table 1 reports the corresponding main-query target comparison.